

## Chapter 6: Pressure

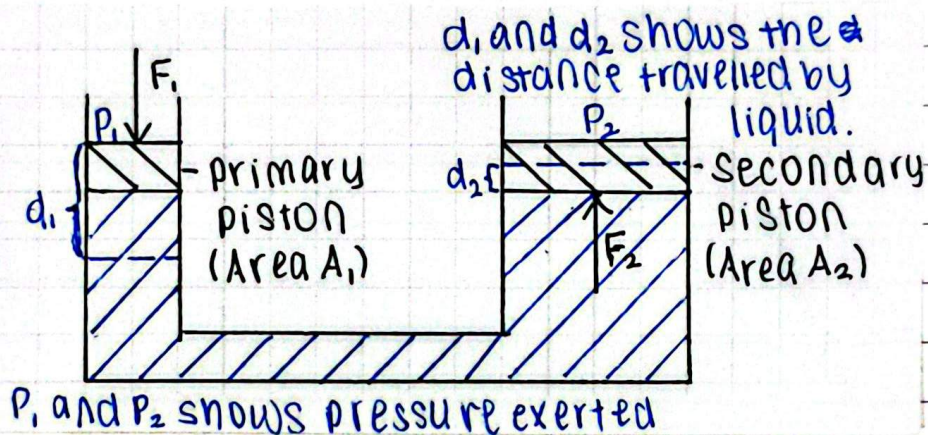
### Pressure (definition)

Pressure is the force acting per unit area of contact.

Pressure = Force / Area (Force is perpendicular to the surface)

- SI unit  $\rightarrow \text{N/m}^2$  or Pascal (Pa)

### Hydraulic Press system



### Pascal's principle (definition)

Pascal's Principle states that if a pressure is applied to an enclosed liquid, the pressure is transmitted to all other parts of the liquid undiminished.

- This is the working principle behind the multiplier effect of the hydraulic press.

### Method 1 of Hydraulic Press System (Force comparison)

By Pascal's principle,  $P_1 = P_2$

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$F_2 = F_1 \left( \frac{A_2}{A_1} \right)$$

Since  $A_2 > A_1$  (i.e.  $\frac{A_2}{A_1} > 1$ ),  $F_2 > F_1$

(Applying a small force at  $F_1$  results in a large force at  $F_2$ )

## Method 2 of Hydraulic Press System (distance comparison)

(Chapter 7 - Energy)

Applying Principle of Conservation of Energy and assuming no friction,

$$\text{work input} = \text{work output}$$

$$F_1 \times d_1 = F_2 \times d_2$$

Since  $F_2 > F_1$ ,  $d_2 < d_1$ .

## Method 3 of Hydraulic Press system (distance comparison)

Since liquid is incompressible, volume of liquid pushed by smaller piston is equal to that pushed by the bigger piston.

Volume of liquid pushed up = Volume of liquid pushed down

$$A_2 d_2 = A_1 d_1$$

Since  $A_2 > A_1$ ,  $d_2 < d_1$ .

Smaller piston travels larger distance than larger piston.

## Density (definition)

Density is defined as mass per unit volume.

$$\text{Density (kg/m}^3\text{)} = \text{mass (kg)} \div \text{volume (m}^3\text{)}$$

SI unit  $\rightarrow \text{kg/m}^3$

In symbols  $\rightarrow \rho = m \div V$

$$1000 \text{ kg/m}^3 = 1 \text{ g/cm}^3$$

## Measuring volume of objects

1. Object has a greater density than the liquid.

Take the reading of liquid,  $V_1$ .

Take the reading of liquid + object,  $V_2$ .

Volume of object  $\rightarrow V_2 - V_1$

2. Object has lesser density than the liquid.  
Find another object that sinks in the liquid.  
Take the reading of liquid + object,  $V_1$ .  
Take the reading of liquid + object of interest tied to the object,  $V_2$ .  
Volume of object of interest  $\rightarrow V_2 - V_1$

Pressure due to a liquid column (definition)

Pressure due to a liquid column at any point  $h$  below the air-liquid surface is the product of height  $h$ , the density of the liquid and the gravitational field strength.

(Pa)                      (m)                      ( $\text{kg/m}^3$ )                      ( $\text{N/kg}$ )  
 $P_{\text{liq}} = \text{Height of liquid column} \times \text{Density of liquid} \times \text{Grav. field strength}$   
 $P_{\text{liq}} = h\rho g$

\* Height measured from surface to point.

Liquid finds its own level

A liquid in an irregularly-shaped container will always find a common level, as the pressure difference due to the different heights will cause the liquid to flow and equalise the pressure difference.

\* Points of similar depth beneath a liquid surface have the same pressure. \*

The larger the depth of a liquid, the greater the liquid pressure.

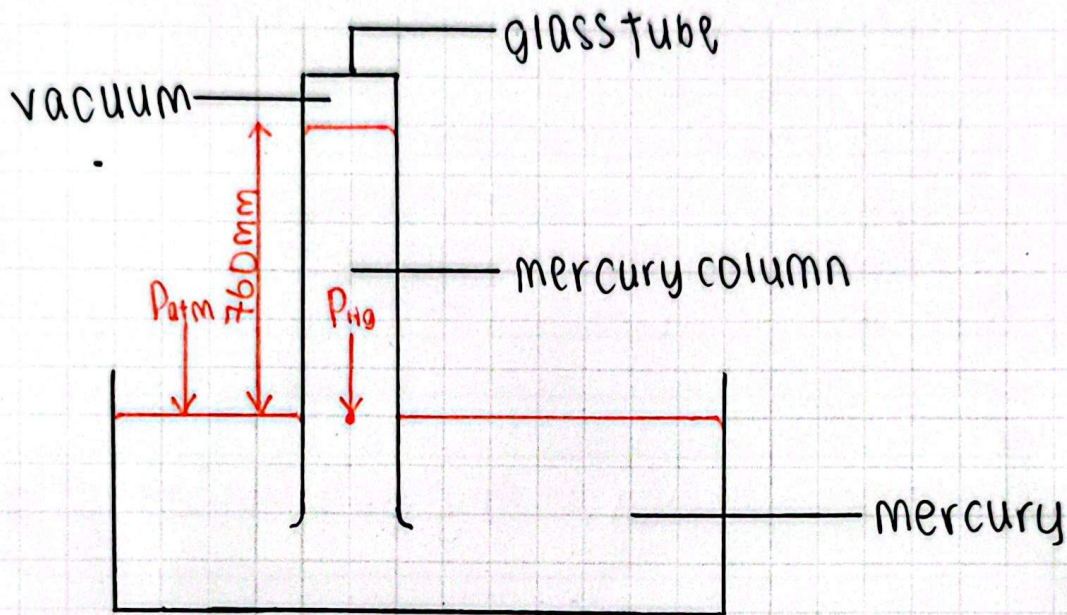
Atmospheric Pressure (definition)

Atmospheric pressure is the amount of weight of air acting per unit surface area that is exposed to the atmosphere. Atmospheric pressure at sea level  $\approx 10^5 \text{ Pa}$ .

## Barometer (definition)

Barometer is a device used to measure atmospheric pressure.

### Mercury Barometer



Since  $P_{atm} = P_{Hg}$ ,  $P_{atm} = 760 \text{ mmHg} \approx 10^5 \text{ Pa}$

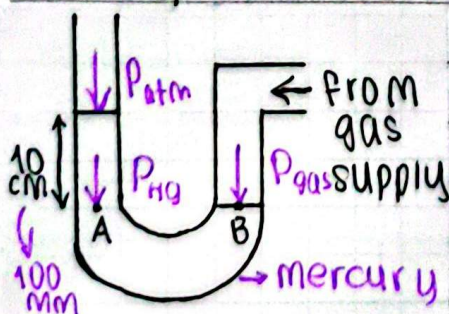
### Convert mmHg to Pa

$$x \text{ mmHg} = \frac{x}{1000} \times 13600 \times 10 \text{ (hpa)} \\ = 136x \text{ Pa}$$

## Manometer (definition)

Manometer is a device used to measure the difference in gas pressure and the atmospheric pressure.

### Finding Gas Pressure using Manometer



Pressure at A = Pressure at B

$$P_{atm} + P_{Hg} = P_{gas}$$

$$760 \text{ mmHg} + 100 \text{ mmHg} = P_{gas}$$

$$P_{gas} = 860 \text{ mmHg}$$

## Chapter 7: Energy

### Energy (definition)

Energy is the capacity of a body to do work.

- SI unit  $\rightarrow$  Joule (J)

- Scalar quantity

### Types of Energy Stores

1. Kinetic store  $\rightarrow$  energy due to motion
2. Gravitational potential store  $\rightarrow$  Energy due to height
3. Chemical potential store  $\rightarrow$  Energy stored in chemical bonds
4. Elastic potential store  $\rightarrow$  Energy in stretched / compressed elastic object
5. Internal store  $\rightarrow$  Energy due to heat
6. Nuclear store  $\rightarrow$  Energy stored in the nucleus of an atom

### Transfer of Energy

1. Energy transferred mechanically by a force acting over a distance (work done)
2. Energy transferred by heating due to temperature difference
3. Energy transferred electrically by an electric current
4. Energy transferred by propagation of electromagnetic waves (infrared, light)
5. Energy transferred by propagation of mechanical sound waves (sound)

### Energy in kinetic store (definition)

Energy in kinetic store is the energy possessed by the object due to its motion.

Formula to calculate energy in kinetic store (in Joules):

$$K.E = \frac{1}{2}mv^2$$

$\rightarrow m$  = mass of object in kg

$\rightarrow v$  = speed of object in m/s

### Energy in Potential store (definition)

Energy in potential store is the stored energy in an object due to its position, shape or state of matter.

### Energy in Gravitational potential store (definition)

Energy in gravitational potential store is the energy which a body possesses due to its height from the ground.

Formula:  $GPE(J) = mgh$ ,

where  $m$  = mass of object (in kg),

$g$  = gravitational field strength (in  $N/kg$  or  $ms^{-2}$ )

$h$  = height from ground (in m)

### Principle of conservation of Energy (definition)

Principle of conservation of energy states that energy cannot be created or destroyed. It can only be transferred from one store to another. The total energy of an isolated system is constant.

In an isolated system, total energy at position A = total energy at position B

### Work Done (definition)

Work done by a constant force on an object is the product of the force and the distance moved by the object in the direction of the force.

SI unit  $\rightarrow$  Joule (J) scalar quantity

Work Done = Force  $\times$  Distance moved in the direction of the force

$W(J) = F(N) \times d(m)$

## Efficiency of an energy transfer

### Efficiency of an energy transfer Formula:

$$\text{Efficiency} = \frac{\text{Useful energy output}}{\text{total energy input}} \times 100\%$$

$$\text{Useful energy output} = \text{efficiency} \times \text{total energy input}$$

## Power (definition)

Power is the rate of work done or rate at which energy is transferred.

SI unit for power  $\rightarrow$  Watt (W)

Formula for power:

$$\text{Power (W)} = \frac{\text{Work done (J)}}{\text{time taken (s)}} \quad \text{OR} \quad \text{Power (W)} = \frac{\text{Energy change (J)}}{\text{time taken (s)}}$$
$$P = \frac{F \times d}{t} \quad \text{OR} \quad P = \frac{E}{t}$$

## Another way to express power

$$\text{Since } P = \frac{F \times d}{t},$$

$$P = F \times \frac{d}{t}$$

$$\text{Power} = \text{Force} \times \text{speed (Distance over Time)}$$

## Power Efficiency

$$\text{Power Efficiency} = \frac{\text{Useful power output}}{\text{total power input}} \times 100\%$$

$$\text{Useful power output} = \text{Power Efficiency} \times \text{Total Power Input}$$

## Energy Resources

### Non-Renewable Energy

→ energy from natural resources that cannot be replenished at a sustainable rate.

→ Fossil Fuel, Nuclear Fuel

### Renewable Energy

→ energy that can be replenished naturally

→ Biofuel, Wind, Tides, Hydropower, Solar, Geothermal

## Non-Renewable Energy Resources

### 1. Fossil Fuel

- Efficiency → High reliability → does not depend on env factors
- Cost → Relatively low cost of production
- Environmental impact → Gases produced during burning of fossil fuels contributes to global warming

### 2. Nuclear Fuel

- Efficiency → High reliability in supplying uninterrupted power
- Cost → High cost to build the infrastructure.  
Additional cost to dispose radioactive waste
- Environmental impact → Does not contribute to global warming  
→ waste produced contaminates groundwater if not disposed properly

## Renewable Energy Resources

### 1. Biofuel

- Efficiency → Highly reliable → widely available
- Cost → Relatively low cost of production.
- Environmental impact → Gases produced during burning contributes to global warming.

## 2. Wind

- Efficiency → Intermittent power. Efficiency depends on wind direction and speed.
- Cost → High land cost
- Environmental Impact → Clean energy, but requires large open areas and possible noise pollution.

## 3. Tides

- Efficiency → More reliable than the wind or sun
- Cost → High cost to build tidal infrastructure.
- Environmental Impact → Clean energy, but affects marine life due to alteration of water current and noise.

## 4. Hydropower

- Efficiency → High efficiency
- Cost → High cost to build infrastructure
- Environmental Impact → Clean energy, but dam construction disrupts ecosystems.

## 5. Geothermal

- Efficiency → Low efficiency
- Cost → High cost to build infrastructure
- Environmental Impact → Poisonous gases such as Hydrogen sulfide released into the atmosphere.

## 6. Solar

- Efficiency → Not very reliable. Depends on weather & availability of sunlight
- Cost → High land cost. Solar panels are expensive.
- Environmental Impact → Clean energy, but takes up space.